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A Appendix

Algorithm 1 Embedding-level Perturbations

Require: Number of attack steps k, ICL examples I with length n_1, unlabeled question x_p with length n_2, ground truth y_{gt}, model f parameterized by θ, attack region ϵ, stepsize α, loss weight λ, embedding layer h, judge LLM J.

Ensure: Correct a, Success b, Unattackable c, Wrong d.

Initialize $a, b, c, d \leftarrow 0, 0, 0, 0$
▷ Initialization for conditions
$(y_p, r_p) \leftarrow \text{argmax}\, P_\theta((y_p, r_p)|I, (x_p))$
▷ Generate original answer and CoT
if $y_p == y_{gt}$ **then**
  $a \leftarrow 1$
  $E_c \leftarrow h(I, (x_p, y_p, r_p))$
  $E_a \leftarrow h(I, (x_p, y_p))$ ▷ Get embeddings
  $q_a, q_c \leftarrow f(E_a), f(E_c)$ ▷ Get logits
  $E'_c \leftarrow E_c$
  for $i \leftarrow 1$ to k **do**
    $q'_c, q'_a \leftarrow f(E'_c), f(E'_c[: n_3])$
    $\mathcal{L}_a, \mathcal{L}_c \leftarrow L_{CE}(q'_a, q_a), L_{CE}(q'_c, q_c)$
    $\mathcal{L}_{opt} \leftarrow \mathcal{L}_c - \lambda \cdot \mathcal{L}_a$ ▷ Loss objective
    $E_{tmp} \leftarrow E'_c + \alpha \cdot \epsilon \cdot \text{sign}(\nabla_{E'_c} \mathcal{L}_{opt})$
    $E_{tmp} \leftarrow \text{clamp}(E_{tmp}, E_c - \epsilon, E_c + \epsilon)$
    $E'_c \leftarrow \text{concat}(E_c[: n_1], E_{tmp}[n_1 : n_2], E_c[n_2 :])$ ▷ Update perturbed embedding
  end for
  $y'_p, r'_p \leftarrow \text{argmax}\, P_\theta((y'_p, r'_p)|E'_c[: n_2])$
  if $y'_p \neq y_{gt}$ **then**
    $d \leftarrow 1$, **Break**
  else
    if $J(r'_p, r_p) == 0$ **then**
      $b \leftarrow 1$, **Break**
    end if
  end if
  if $i == k$ **then**
    $c \leftarrow 1$
  end if
end if
return a, b, c, d ▷ Final conditions

A.1 Limitations

Our method requires significant computational resources due to the complexity of generating and evaluating successful examples on LLMs. The effectiveness of our approach relies on the LLM judge's ability to assess the correctness of reasoning within CoT explanations. However, this evaluation process is not entirely reliable, as LLMs may exhibit inconsistencies or biases in their judgments, which could potentially affect the accuracy of our robustness analysis. Further, the token-level attack may slightly impact the original meaning of the questions.

A.2 Details of datasets

SingleEq (Koncel-Kedziorski et al., 2016) is a collection of algebraic word problems designed to evaluate the mathematical reasoning capabilities of machine learning models. Each problem in the dataset consists